

When deciding upon the direction that this project would head in I wanted to ensure that my work on this would result in me learning something new that wasn't touched on in class. From there I decided to examine the Plotly library to see if there were any visualizations that caught my eye as something that is both practical in use and new material for me to learn. I landed on the idea of creating a tutorial on the choropleth map which is a type of map that has shaded regions to represent numeric values. While we touched briefly on the basics of these types of visualizations, I wanted to experiment with making a choropleth map through Plotly, which allows for the creation of interactive graphs. Choropleth maps are often used when showing differences across regions, making this a popular visualization seen often on the news and online. This popularity separated this visualization from the other great options in Plotly's library, leading me to ultimately decide upon creating my tutorial about choropleth maps.

The tutorial that I developed draws largely on the content that I learned in class from both the readings and the discussions that we had. The idea of demonstrating all the different attributes to the plot was something that I felt was important after considering all the different visualization types and purposes that we learned about in our readings. Readings like Tufte discussed the importance of using the correct data visualization for the data and ensuring that the data is presented in the clearest way possible. Having the ability, and understanding, to alter all the different attributes of the visualization allows for clarity to be ensured. Understanding the possibilities of the attributes of the plot also allows for adjusting the plots in a way to create an argument about the data. Overall, many of the decisions that were made in respect to the direction of this project were made with the teachings of this course in mind.

The largest struggle that I had developing this tutorial was deciding upon which type of visualization that I wanted to focus upon, and then from there deciding which type of choropleth graphing package would be the best to use. While this may not seem like a very big struggle, it was an important decision that I spent a lot of time considering, weighing my options to ensure that I chose the best options for my tutorial.

The part of the tutorial that I am the most proud of would have to be the number of different attributes that are available to be examined and demonstrated within the tutorial. I worked through all the different attributes of the choropleth graph that were applicable for the dataset I was using, and gave a demonstration of how it could be used to alter the final visualization. I am proud of the way that this tutorial allows for customization of the algorithm and that it demonstrates the many uses of the choropleth mapping package within Plotly.

If I had more time I would work on adding the animation aspects to the tutorial. There were many interesting things that could be done by adding extra variables, and then creating animations and widgets. While it would have been interesting to add these aspects to the tutorial that I created, there just wasn't enough time to reasonably complete this.